

Chapter I

Introduction to Databases

- Lecture slides are to be used as reference material only. They are not a substitute of the textbook, lectures and class participation.
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Discussion Objectives

1. What is a Database?
2. File-Based Systems
3. What is a Database Management System (DBMS)?
4. Functions of a DBMS
5. Components of a DBMS
6. Personnel involved in the DBMS environment

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We are generating data

- ❑ Purchases from the supermarket
- ❑ Purchases using your credit card
- ❑ Booking a holiday at the travel agents
- ❑ Using the local library
- ❑ Buying insurance
- ❑ Using the Internet
- ❑ Student records at a university

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What is a Database?

A database is a

- ❑ Computerized **record-keeping system**—an electronic filing cabinet.
- ❑ A collection of **computerized data files** (i.e. data records). Users can *add*, *retrieve*, *modify* and *delete* data in a database.
- ❑ Shared collection of **logically related data** (and a description of this data), designed to meet the information needs of an organization.

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Data in a Database

A database contains logically related data that comprises:

Entities: A distinct real-world object/concept that need to be represented in a database, e.g. person: [patient, nurse], place, thing: [computers, house, car] , event

Attributes: A property that describes some aspect of the object that we wish to record, eg. Person: [Name, Age, Address]

Relationships : An association between entities, eg. doctor *treats* patients, person *drives* a car

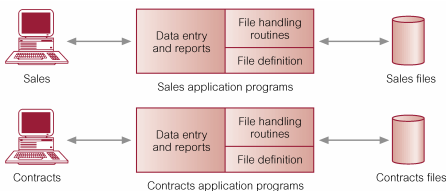
File-based Systems: An Earlier Approach

- ❑ Early attempt to computerize manual filing systems

Collection of application programs that perform services for the end users (e.g. reports).

- ❑ Each program defines and manages its own data.

File-based Processing



Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

PrivateOwner (ownerNo, fName, lName, address, telNo)

Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)

PropertyForRent (propertyNo, street, city, postcode, rent)

Client (clientNo, fName, lName, address, telNo)

Limitations of File-based Approach

- **Separation and isolation of data**
 - Each program maintains its own set of data.
 - Users of one program may be unaware of potentially useful data held by other programs.
- **Duplication of data**
 - Same data is held by different programs.
 - Wasted space and potentially different values and/or different formats for the same item.

Limitations of File-based Approach

- **Data dependence**
 - File structure is defined in the program code.
- **Incompatible file formats**
 - Programs are written in different languages, and so cannot easily access each others files.
- **Fixed Queries/Proliferation of application programs**
 - Programs are written to satisfy particular functions. Any new requirement needs a new program.

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Database Approach

- **Motivation:**
 - Data definition was embedded in application programs, rather than being stored separately and independently.
 - No control over access and manipulation of data beyond that imposed by a single application program.
- **Result**
 - Database and Database Management System (DBMS).

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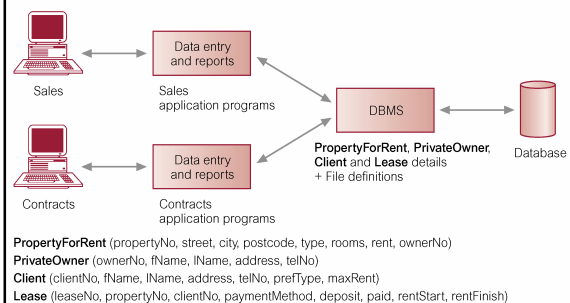
Database Management System (DBMS)

- A software system that enables users to **define, create, and maintain the database**
- A software system which provides **controlled access to the database.**
- **DBMS is not about data storage it is about the management of the data storage!**

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Database Management System (DBMS)



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Components of a DBMS Environment

❑ Hardware

- Can range from a PC to a network of computers.

❑ Software

- DBMS, operating system, network software (if necessary) and also the application programs.

❑ Data

- Used by the organization and a description of this data called the **schema**.

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Components of DBMS Environment

❑ Procedures

- Instructions and rules that should be applied to the design and use of the database and DBMS.

❑ People

- Data Administrator (DA)
- Database Administrator (DBA)
- Database Designers (Logical and Physical)
- Application Programmers
- End Users (naive to sophisticated)

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DBMS: Database Access

❑ Controlled access to database involves a view mechanism

- » users view only the data they want or need to use.

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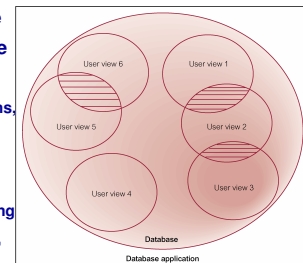
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Views

A view is essentially some subset of the database.

❑ Allows each user to have his or her own view of the database.

- ❑ Patient view (medications, appointments, history, resources, accounts)
- ❑ Physician view (tasks, resources, patients, billing)
- ❑ Manager view (accounts, resources, schedules)



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User Views

User: Student

COURSE Table				
Course ID	Title	Credits	Pre-requisites	
CS123	Intro. to Computers	2	---	
CS234	Databases	3	CS123	
CS345	Networking	4	CS256	
CS456	AI	3	CS367	

User: Staff

STUDENT Table					
BannerID	LName	Major	T. Credits	GPA	Course
B00125	Smith	CS	24	2.78	CS234
B00134	Dole	Arts	56	3.34	AR354
B00345	Ahmad	Law	36	2.35	LA246
B00343	Kumar	Cs	96	3.22	CS456
B00657	Ying	Music	68	2.86	MU433

FACULTY Table				
BannerID	LName	Faculty	Level	Teaches
B006255	Richard	CS	Prof.	CS123
B006345	Brown	Science	Assoc. P	AR354
B007455	Ali	Law	Prof.	LA246
B002436	Mehra	CS	Assoc. P.	CS234
B006578	Cheah	Engg.	Assis. P	MU433

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User Views

□ Benefits include:

- Reduce complexity;
- Provide a level of security;
- Provide a mechanism to customize the appearance of the database;
- Present a consistent, unchanging picture of the structure of the database, even if the underlying database is changed.

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The End
Chapter 1

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